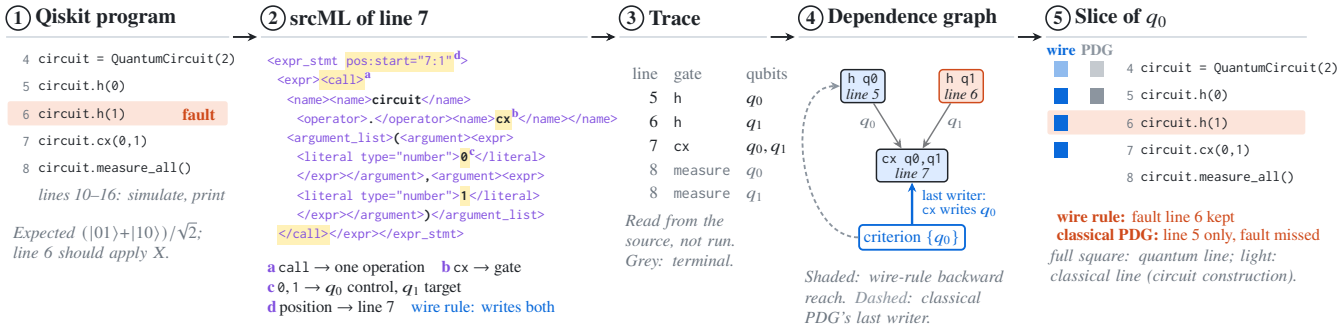

1 Qiskit program
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1) fault
7 circuit.cx(0,1)
8 circuit.measure_all()
lines 10–16: simulate, print
Expected (|01⟩+|10⟩)/√2; line 6 should apply X.
2 srcML of line 7
<expr_stmt pos:start="7:1"d>
<expr><call>a
<name><name>circuit</name>
<operator>.</operator><name>cx</name>b</name></name>
<argument_list>(<argument><expr>
<literal type="number">0c</literal>
</expr></argument>,<argument><expr>
<literal type="number">1</literal>
</expr></argument>)</argument_list>
</call></expr></expr_stmt>
a call → one operation
b cx → gate
c 0, 1 → q0 control, q1 target
d position → line 7
wire rule: writes both
3 Trace
line gate qubits
5 h q0
6 h q1
7 cx q0, q1
8 measure q0
8 measure q1
Read from the source, not run.
Grey: terminal.
4 Dependence graph
h q0
line 5
h q1
line 6
q0
q1
cx q0,q1
line 7
last writer:
cx writes q0
criterion {q0}
Shaded: wire-rule backward reach. Dashed: classical PDG's last writer.
5 Slice of q0
wire PDG
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)
7 circuit.cx(0,1)
8 circuit.measure_all()
wire rule: fault line 6 kept
classical PDG: line 5 only, fault missed
full square: quantum line; light: classical line (circuit construction).